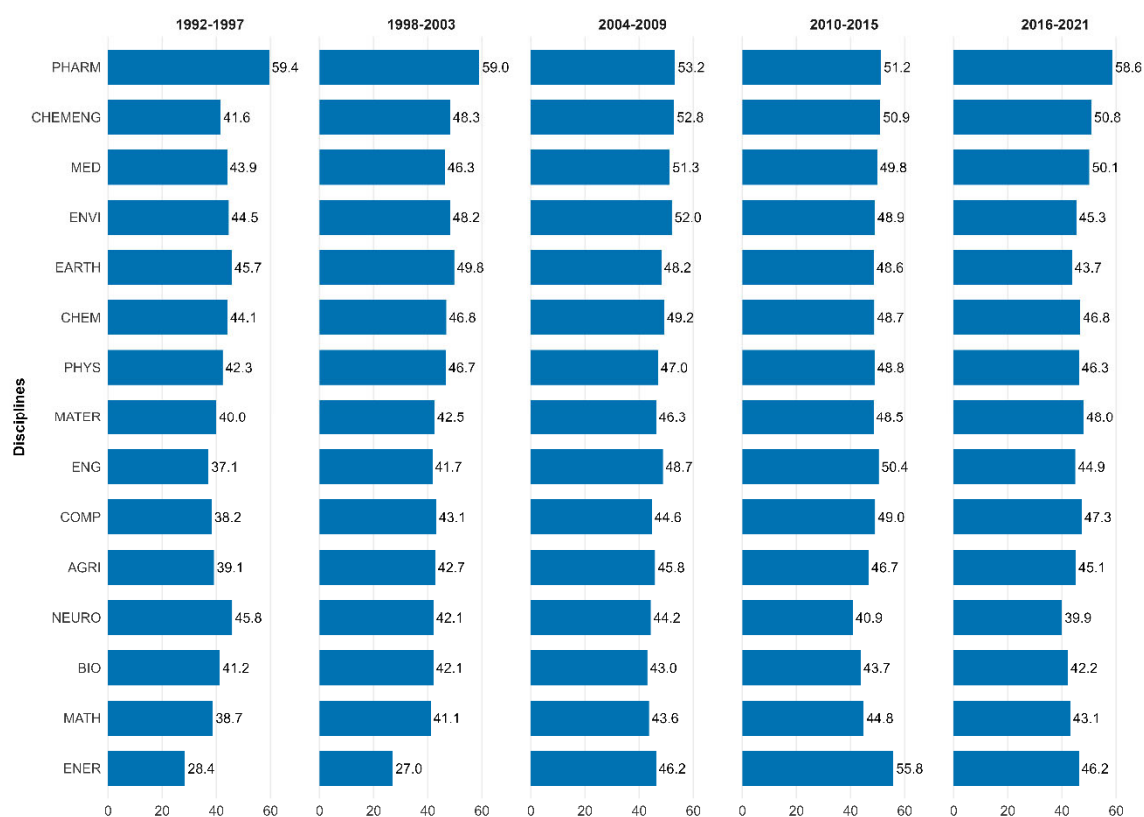


Figure 3 shows the percentages of total journal prestige-normalized production by the authors belonging to the top performer class, broken down by discipline and five consecutive time periods from 1992–1997 to 2016–2021. Each bar represents the percentage share of production generated by the top 10% of authors in a given period $\times$ discipline cell, thus illustrating the scale of production concentration measured not only by the number of publication, but also by the journal prestige. In the construction of journal prestige-normalized production, journal prestige is taken into account in a nonlinear manner: the weight of publications increases with the percentile of the journal according to the power of 2.5 (the Scopus percentile ranks ranging from zero to one), which means that works published in the highest segment of the journal ranking have a disproportionately large impact on the total value of the indicator.

Figure 3. Production of the 10% top-performing group in terms of nonlinear journal prestige-normalized productivity ($x^{2.5}$)



A strong concentration of research production is visible at first glance. In all disciplines and in all periods, the top decile of authors generates a significant portion of total knowledge production – usually from about one-third to more than half. This level of concentration means that production, taking into account the prestige of publication channels, is highly concentrated in a relatively narrow class of researchers (the 10/50 rule examined in Kwiek & Roszka, 2024: 10% of scientists in the Polish science system are responsible for 50% of national knowledge production). The nonlinear weighting approach to productivity means that a moderate advantage in access to more prestigious journals translates into significantly higher productivity (in a linear approach, Scopus journal percentile ranks are used according to the formula $y=x$, in our nonlinear approach, the rank are used according to the formula $(y = x^{2.5})$).

The scale of concentration varies between disciplines, but the pattern itself remains stable. The highest percentages of publications produced by top performers are consistently observed in areas

such as PHARM, CHEMENG, MED, ENVI, and PHYS. On these disciplines, the share of production by the top decile of authors often exceeds 50%). These are also disciplines in which publishing in the highest percentile journals plays a particularly important role. In disciplines such as MATH, COMP, and AGRI, production is distributed more evenly, although even there, it remains clearly concentrated in the hands of a small proportion of authors, especially in later periods.

There has been an increase in the concentration of production from the 1990s to the mid-2000s, followed by stabilization at a high level. In some fields, including ENG, ENER, and PHYS, the share of total production by top performers increased after 2010. This suggests that the advantage of the top decile of authors not only persisted but was strengthened in some areas.

It is worth noting that even in the disciplines where the share of production by top performers is lowest, productivity is not evenly distributed. Although authors outside the top performer class generate the majority of publications in purely quantitative terms, these are largely works published in less prestigious journals. Due to the nonlinear weighting of prestige, their total contribution to productivity therefore remains clearly limited.

Overall, the concentration of research production in Polish science is structural and stable over time. Top performers both publish more and dominate the segment of highly prestigious publications, which (given the weighting method used) translates into a disproportionately large share of total production. This result complements earlier analyses and confirms that stratification by productivity is one of the key mechanisms organizing access to prestigious journals.

Figure 4 shows the distribution of articles by Scopus journal deciles (expressed as percentile ranks), broken down into the top performer class and the remaining scientists. The analysis is conducted separately for disciplines and subsequent time periods. The horizontal axis shows journal deciles (from 0–9 to 90–99), while the vertical axis shows the share of articles published in a given quality segment. The lines show the average shares for the two classes of scientists and allow for a direct comparison of their publication profiles across the entire quality scale.

In almost all disciplines and periods, a clear quality gradient is visible. The share of articles systematically increases with the transition to higher journal deciles. The highest concentration of publications appears in the three deciles of 70–79, 80–89, and 90–99. Against this background, the differences between top performers and the remaining scientists are most evident in the highest deciles. The curves for top performers are consistently shifted upward relative to other scientists in the 90–99 percentile segment. In many disciplines this difference begins to be visible in the 70–79 and 80–89 deciles. In later periods, this shift becomes more pronounced, indicating that the advantage of top performers comes from access to prestigious journals.

In the lower and middle parts of the quality distribution (up to about the 40–49 percentile), the publication profiles of both groups are very similar, and in many cases they almost overlap. This suggests that the differentiation between top performers and the rest is not about publishing “more everywhere” but only manifests itself at the top of the journal hierarchy.

Despite noticeable differences between disciplines – evident, for example, in the different publication profiles in PHYS, MED, COMP, and ENER – the relationship between top performers and the remaining scientists remains surprisingly stable. Regardless of discipline, the advantage of top performers is small in the lower deciles and clear in the highest ones, which suggests the existence of a common selection mechanism for the publishing elite.

In dynamic terms, a gradual intensification of concentration in the highest deciles is visible after 2010, especially in 2016–2021. This observation is consistent with earlier findings regarding the growing share of publications in the 90–99 percentile segment and the increasing concentration of production as measured by the journal prestige-normalized productivity index in the top performer class.

Overall, Figure 4 shows that the differences between top performers and other scientists are primarily qualitative rather than quantitative. Top performers do not publish more frequently across the entire spectrum of journals but disproportionately publish in the highest rated publication channels. At the same time, the similarity of distributions in the lower deciles indicates that the stratification of science is mainly evident at the top of the journal hierarchy, rather than in its lower segments.

Figure 5 shows the difference between article percentages between top performers and the remaining authors in successive deciles of Scopus journals. The analysis is conducted separately for disciplines and for five periods from 1992–1997 to 2016–2021. Each cell of the heat map represents the value

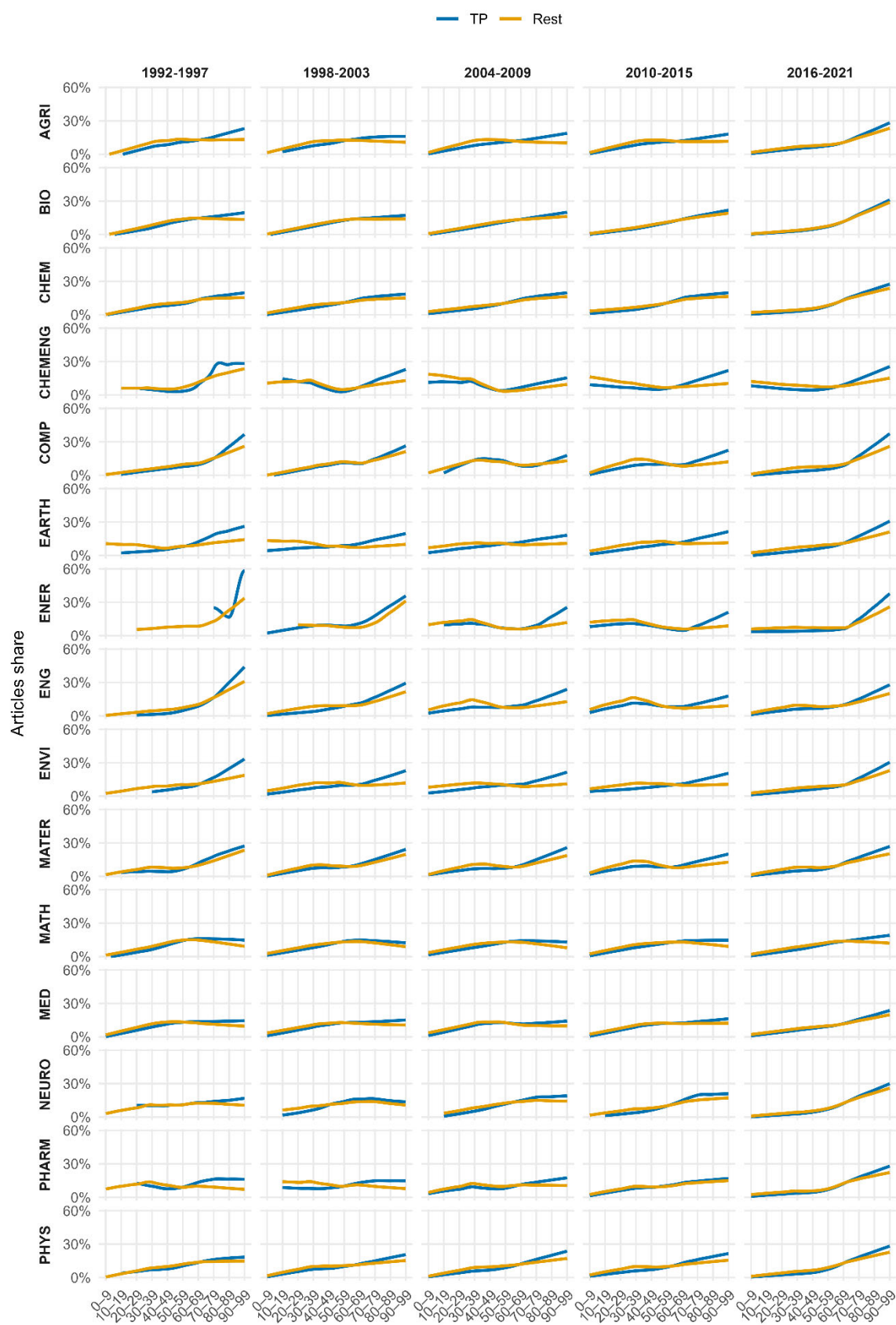
$$\Delta = share_{TP} - share_{Rest},$$

where darker shades indicate a top performer (*TP*) advantage, and light shades indicate advantage for the rest (crosses indicate no or too few observations).

The plot reveals a regular pattern of changes in the sign of differences along the journal quality scale. In the lower and middle deciles of journal ranks (up to about the 40–49 percentiles), differences are close to zero or negative. This means that top performers do not dominate in publishing in low-quality journals. In these segments, the publication profiles of both groups are very similar. It is different in deciles 70–79, 80–89, and 90–99: the differences become positive and often exceed 20 percentage points. This means that the top performer advantage is concentrated almost exclusively at the top of the journal hierarchy. And that the advantage manifests itself in the form of overrepresentation of publications in the most prestigious channels.

In dynamic terms, a change in the intensity of this pattern is observable. In the earliest period (1992–1997), positive values in the highest deciles are present. In subsequent periods, especially after 2010, they form distinct dark bands in the 90–99 percentile segment. This indicates a gradual strengthening of the concentration of publication quality in the top performer class.

Figure 4. Article average percentages in journal deciles, top performers (TP) vs. other scientists. All deciles = 100%.



Despite clear differences between disciplines, visible, among others, in different profiles for PHYS, CHEM, MED, BIO, ENG, ENER, and COMP, the relationship between top performers and the other scientists remains surprisingly consistent. Regardless of the discipline, positive differences appear systematically only in the highest quality deciles, while in the lower segments of the distribution of journal prestige they are weak or absent.

The pattern indicates that the advantage of top performers does not lie in publishing “more everywhere” but in selective access to the most prestigious journals. The stratification of scientific production is therefore clearly qualitative in nature: the differences between top performers and other authors are revealed almost exclusively at the top of the publishing hierarchy, and their intensity has increased over time.

These findings lead to the conclusion that inequalities in science are generated primarily by two mechanisms: prestige concentration and selection into the publishing elite. They are not produced by uniform differences across the entire publication spectrum.

Figure 5. Difference in shares by top performers (TP) and the rest of scientists, by journal decile, period, and discipline

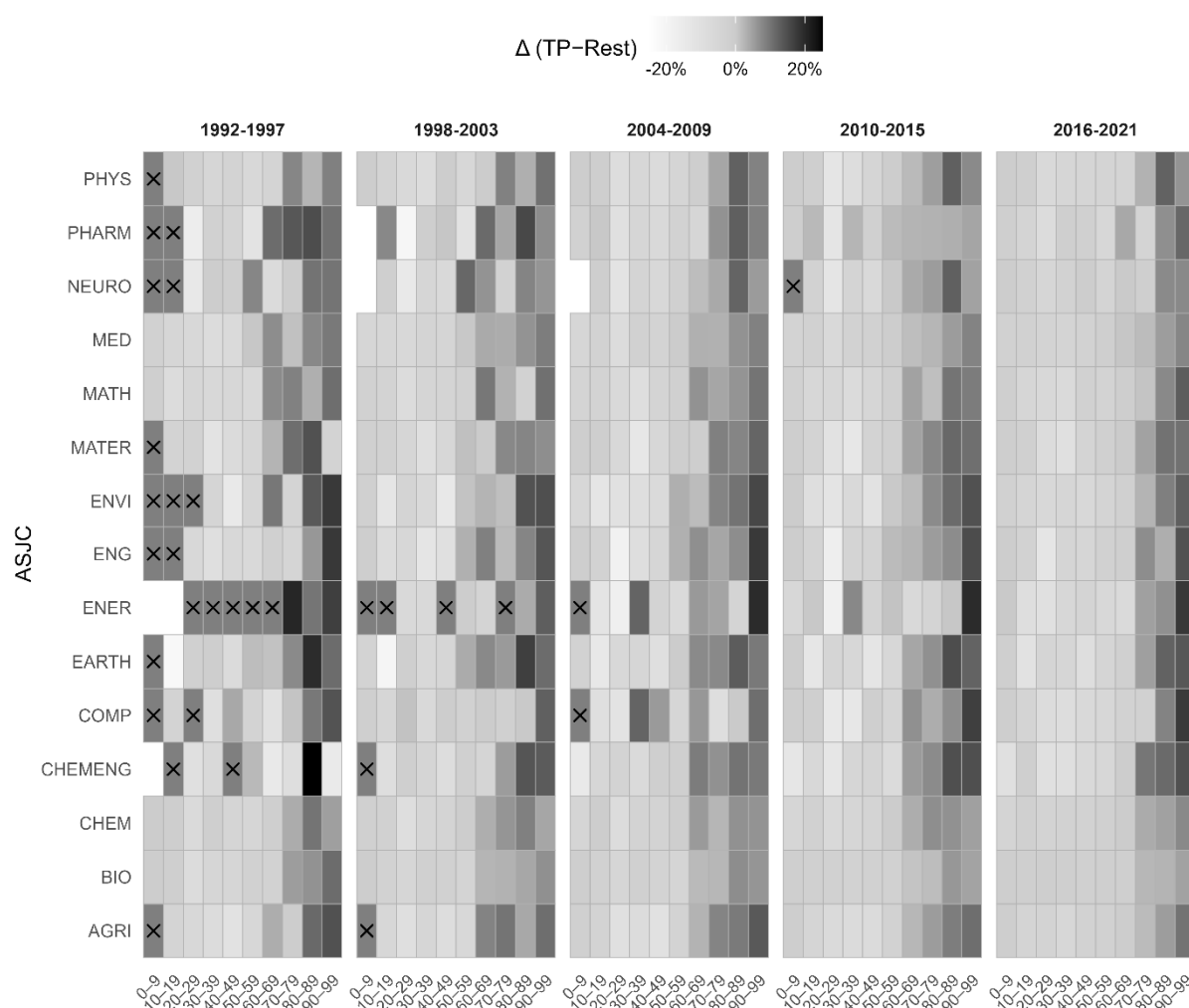


Figure 6 shows the average distribution of articles by Scopus journal decile for the top and other authors, separately for five consecutive periods, for all disciplines combined. Each curve shows the